

Recovering the Unrecoverable

Study H-R — the 12 Study H injections that found no trajectory, re-solved with the state corridor relaxed

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1 The question

Study H injected 12-odd faults along an Apollo descent and found that **11** of its injections had no recovery trajectory at all: IPOPT declared local infeasibility from seven independent seeds at up to 1,200 iterations. The natural reading of that result is “the vehicle is lost”. It is not the only reading.

The optimal-control problem those solves failed carries a corridor of path constraints, and **not all of them are physics**:

Constraint	Study H value	What it actually is
Glide cone	12° above horizontal	A scenario choice. It exists to rule out the dive-and-crawl-back trajectory family, not because a vehicle cannot fly below it.
Attitude	45° per axis	Planner comfort. <code>fault_lib</code> ’s own <code>hard_loss()</code> puts genuine loss of control at 90°, twice this.
Body rate	10°/s	Comfort again; <code>hard_loss()</code> uses 120°/s.
Speed	60 m/s per axis and as a norm	The vehicle model has no aerodynamic or structural speed limit at all; this is a modelling convenience.

So “no trajectory exists” may mean “no trajectory exists **that stays inside the corridor**”. Those are very different engineering statements. The first is a lost vehicle. The second is a vehicle that could be saved by a

guidance law willing to leave the nominal envelope — and knowing which one you have is worth more than the failure count itself.

This study separates them. It re-solves every case Study H could not land, at progressively weaker corridors, and records **the weakest relaxation that admits a trajectory** together with **how far outside the baseline limits that trajectory actually goes**.

2 Method

2.1 The ladder

Each case is re-attacked at 5 levels, stopping at the first that yields a trajectory:

Level	Corridor	Constraints removed outright
L1	glide cone 12 to 6 deg	none dropped
L2	+ attitude 45 to 60 deg, rate 10 to 20 deg/s	none dropped
L3	+ speed 60 to 90 m/s (axis and norm)	none dropped
L4	corridor removed (cone, speed, attitude, rate)	cone, vel, att, rate
L5	all state constraints removed, altitude floor included	cone, vel, att, rate, alt

2.2 What is never relaxed

The ladder changes what the vehicle is allowed to do **en route**. It never changes what counts as success, and it never invents hardware:

- **Thrust and gimbal bounds stand at every level.** They are actuator hardware; relaxing them would answer a question about a vehicle that does not exist.
- **The Apollo landing gate stands at every level.** A relaxed solve still has to touch down inside the same gate to be counted as a landing, which is why the results below distinguish *found a trajectory* from *landed*.
- **The altitude floor stands through L4**, and is dropped only at **L5**.

L5 needs stating plainly, because it is not a corridor at all: with the floor gone there is **no constraint on the state left anywhere in the problem** — only the dynamics and the actuator bounds. A trajectory found there may pass through the lunar surface, and it is therefore *not a landing* and is not counted as one. Its only job is diagnostic: it separates “the state constraints made this infeasible” from “this plant cannot fly this manoeuvre at all”. A case that fails at L5 has failed on physics no relaxation can reach.

2.3 Effort parity

A relaxed solve that fails must fail for the same reason the baseline did, not because it was tried less hard. Study H spent 7 seeds at up to 1,200 iterations on each of these cases (2 in the campaign, 6 in the hardening pass). Every level here gets 3 horizon seeds at 1,200 iterations, so a case still infeasible at L5 has now resisted **22 seeds** in total.

3 Results

Of the 12 cases carried forward — 11 that found no trajectory and 1 that flew but missed the gate — **4 found a trajectory** once the corridor was relaxed, and **2** of those still land inside the unchanged Apollo gate. 1 of the trajectories was found only at L5, with every state constraint gone including the surface; it flies below the ground and is therefore reported as **subsurface** rather than counted as a landing. **8** remained infeasible at every level.

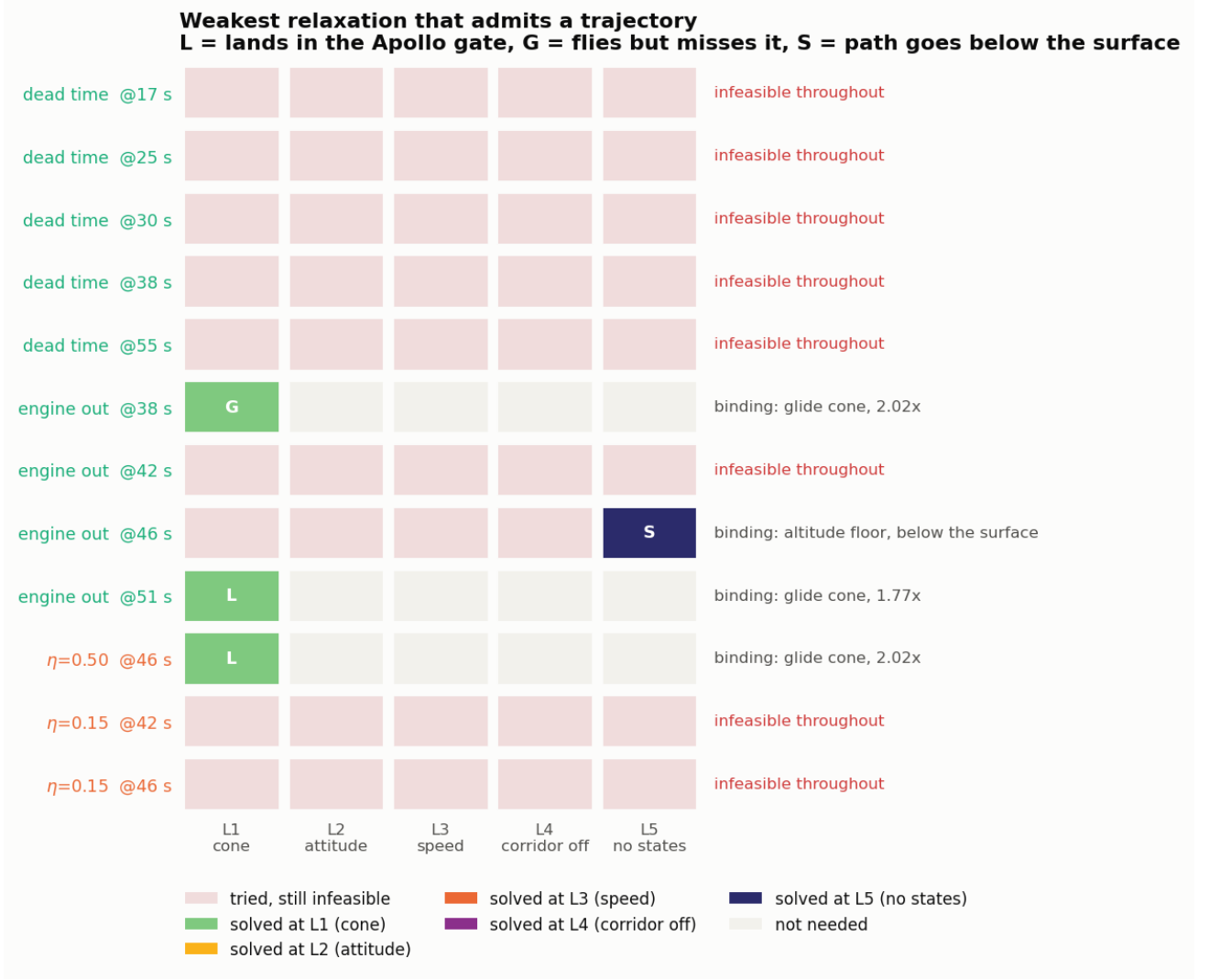


Figure 1: The weakest relaxation that admits a trajectory

Fault	Injected	Altitude	Study H	Solved at	Outcome	Gate margin	Binding baseline limit	Exceeded by
dead time	17 s	419 m	no recovery	—	no recovery	—	n/a	nan×
dead time	25 s	282 m	no recovery	—	no recovery	—	n/a	nan×
dead time	30 s	212 m	no recovery	—	no recovery	—	n/a	nan×
dead time	38 s	109 m	no recovery	—	no recovery	—	n/a	nan×
dead time	55 s	8 m	no recovery	—	no recovery	—	n/a	nan×
engine out	38 s	109 m	gate miss	L1	gate miss	6.88	glide cone	2.02×
engine out	42 s	56 m	no recovery	—	no recovery	—	n/a	nan×
engine out	46 s	19 m	no recovery	L5	subsurface	0.60	altitude floor	inf×

Fault	Injected	Altitude	Study H	Solved at	Outcome	Gate margin	Binding baseline limit	Exceeded by
engine out	51 s	4 m	no recovery	L1	land	0.60	glide cone	1.77×
$\eta=0.50$	46 s	19 m	no recovery	L1	land	0.60	glide cone	2.02×
$\eta=0.15$	42 s	56 m	no recovery	—	no recovery	—	n/a	nan×
$\eta=0.15$	46 s	19 m	no recovery	—	no recovery	—	n/a	nan×

Read the last two columns first. “Binding baseline limit” is the Study H constraint the relaxed trajectory leans on hardest, and “exceeded by” is the peak excursion as a multiple of that limit. A value of $1.00\times$ would mean the relaxation was never used — the trajectory stayed inside the original corridor and only the *solver* needed help. Anything above $1.00\times$ is a constraint the recovery genuinely could not have respected.

3.1 Which constraint was actually binding

Binding limit	Cases	Which
glide cone	3	engine_out@38, engine_out@51, thrust_loss_50@46
altitude floor	1	engine_out@46

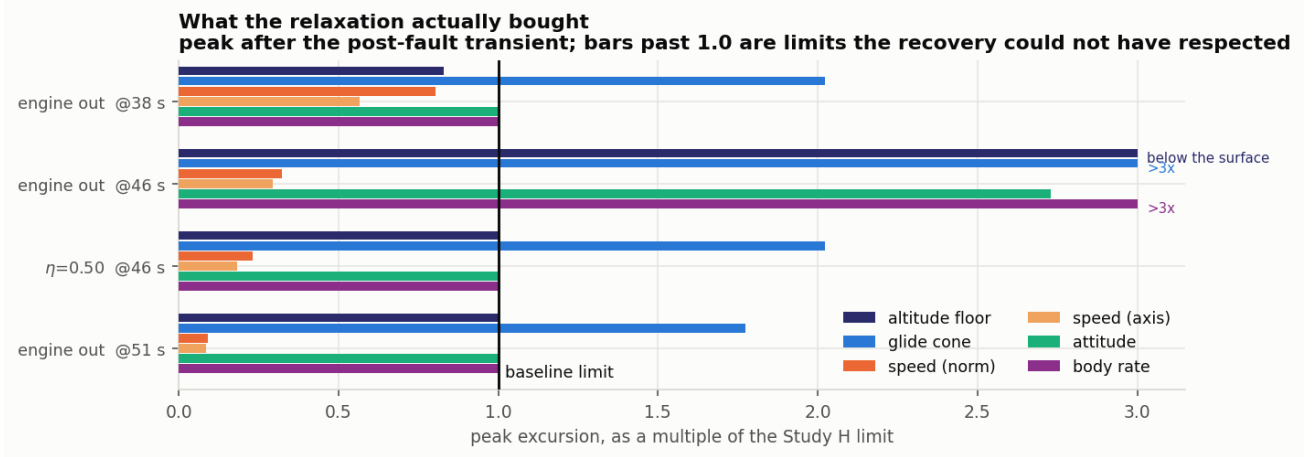


Figure 2: Peak excursion beyond each baseline limit, per recovered case

These excursions are measured after the post-fault transient, from node 6 onward. `solve_ocp` opens a corridor over the first 6 nodes — up to 3x the rate limit, 1.35x the attitude limit, 1.2x the speed limit — which shrinks to the nominal envelope by node 6 and which the baseline problem opens too. An excursion inside that window is something Study H already permitted, not something the relaxation bought, so counting it would credit the recovery to the wrong constraint. Measured over the whole trajectory, two of the recoveries here look like they were bought by the body-rate limit at $\sim 2x$; measured after the ramp, their rate and attitude peaks sit at exactly $1.00x$ and the glide cone is the only limit genuinely broken — which is the answer that agrees with all three having solved at L1, where the cone is the only thing that moved.

3.2 The trajectories themselves

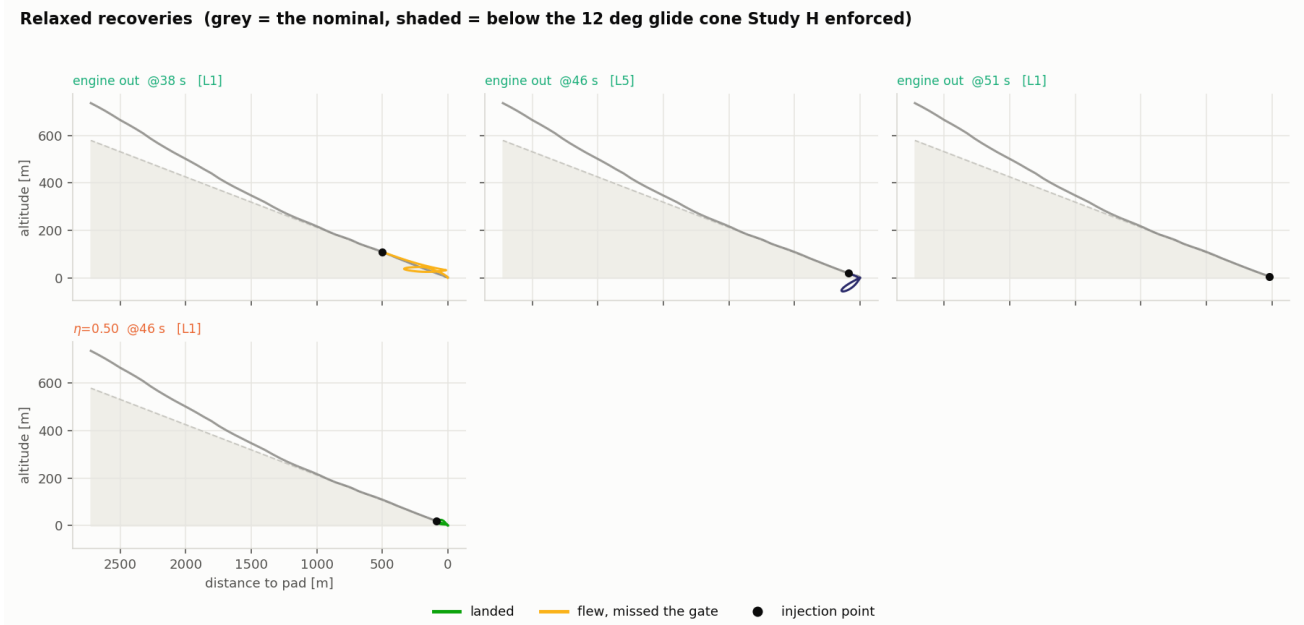


Figure 3: The relaxed recoveries against the nominal and the original cone

4 What this means

The surface is where the diagnosis splits. 1 of the cases have a trajectory only once the altitude floor is removed, diving as deep as 57 m below it. That is not a recovery — the vehicle reaches the pad by flying through the ground — and the touchdown score does not reveal it, because the gate inspects only the final state: this trajectory scores a clean margin at the end of a path that spent most of its length underground. It is reported as **subsurface**, not as a landing. What it establishes is diagnostic: the plant retains enough authority to make the *geometry*, and what it cannot do is make it while staying above the surface. A case that fails even there is short of control authority outright.

A failed solve is a statement about the problem, not only about the vehicle. 4 of 12 cases that Study H recorded as unrecoverable have a trajectory once the corridor is widened. Reporting them as lost vehicles would have overstated the fault’s severity by exactly that much.

The relaxation is not free, and the excursion column prices it. Every recovered case names a specific limit it had to break and the factor by which it broke it. That is the number a guidance designer needs: not “relax the constraints” but “this fault is survivable if the vehicle is permitted altitude floor; glide cone beyond the nominal envelope”.

The gate is the honest discriminator. 2 of the 4 recovered cases land inside the Apollo gate; the rest fly a trajectory to the surface but arrive outside it. A trajectory that exists is not the same as a landing, and the ladder deliberately keeps those separate.

8 cases did not yield to any relaxation (dead time @17 s, dead time @25 s, dead time @30 s, dead time @38 s, dead time @55 s, engine out @42 s, $\eta=0.15$ @42 s, $\eta=0.15$ @46 s). With the entire corridor removed, the only constraints left are the actuator bounds, the surface, and the dynamics — so these are the cases where the *vehicle* genuinely cannot get to the pad, not the cases where the planner was being asked to fly politely. They are the ones a fault-tolerance budget should carry as real losses.

5 Limitations

- **Local infeasibility is still not global.** 22 seeds is strong evidence, not a proof. A case that resists the whole ladder has resisted a lot of search; it has not been shown impossible.
- **The relaxed corridors are chosen, not derived.** 6°, 60°, 20°/s and 90 m/s are round numbers picked to bracket the baseline, not limits computed from vehicle structure or sensor range. The ladder measures *which* constraint binds and *roughly how much* is needed, not the exact threshold.
- **Relaxing a planner constraint does not make the vehicle able to fly it.** A trajectory that pitches past 45° is admissible here because the model has no structural or plume-impingement limit at that attitude. A real vehicle may.
- **Everything inherited from Study H still applies:** instantaneous perfect detection, open loop, one nominal, constant mass.

6 Reproducing

```
python studies/fault_injection/run_relaxed_study.py      # the ladder campaign
python studies/fault_injection/analyse_relaxed.py        # figures + headline JSON
python studies/fault_injection/build_relaxed_report.py  # this document
```